PHD COURSE · AI- AND DATA-DRIVEN DRUG DESIGN

From folding to co-folding to design

Lecture 7 · Deep learning for protein structure and protein design



CPDSE

Albert J. Kooistra

Data Science for Drug Design, University of Copenhagen

Day 3 · 11 November 2026

ROADMAP

Three questions, in order

1

Protein folding

Can we predict the structure of a single chain from its sequence?

2

Protein co-folding

Can we predict what two or more molecules look like together?

3

Protein design

Can we invert the problem and build a protein that binds?

SCORING METRICS

SEGMENT 1

Protein folding

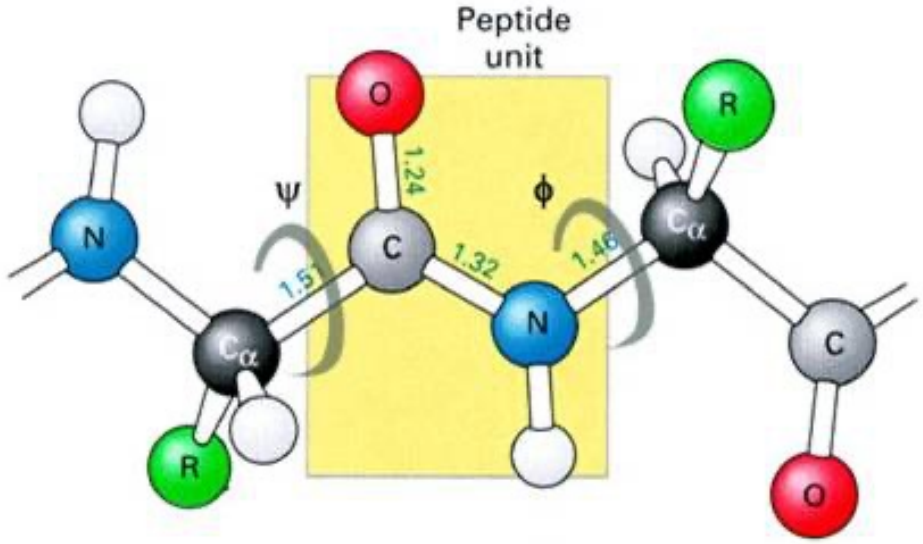
From an impossible search problem to a solved benchmark in roughly fifty years.

THE PROBLEM

Levinthal's paradox

Small protein

- 100 amino acids
- 99 peptide bonds
- 198 phi/psi angles (neglecting omega)
  - Assume 3 stable conformations
- 3^{198} possible conformations



Let’s say 0.1 picosecond per conformation

THE PROBLEM

Levinthal's paradox

$3^{198} = 10^{47}$

conformations for a 100-residue chain

$\sim 10^{27}$ years

to sample them all at 10^{-13} s per state

µs to s

actual folding time observed in the lab

A protein cannot be searching conformational space at random.

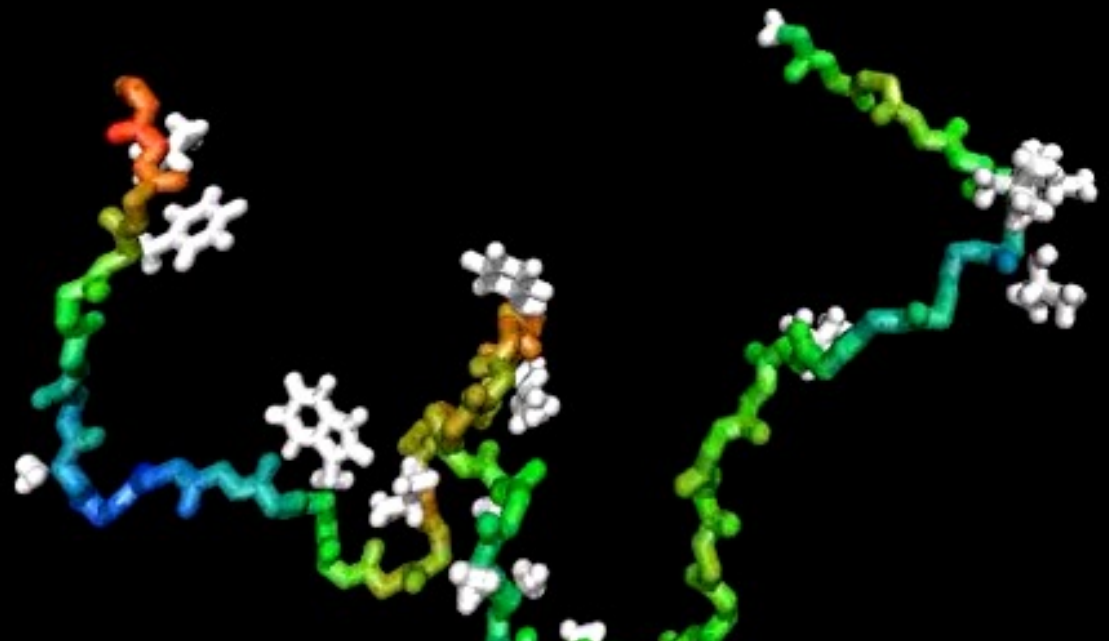
Anfinsen (1972): the sequence contains all the information needed to specify the native structure.

Levinthal (1969): the structure is therefore reachable, but not by enumeration. Folding has to be guided.

The practical consequence: if folding is guided by regularities, those regularities can in principle be learned.

Physic—based Protein folding in action

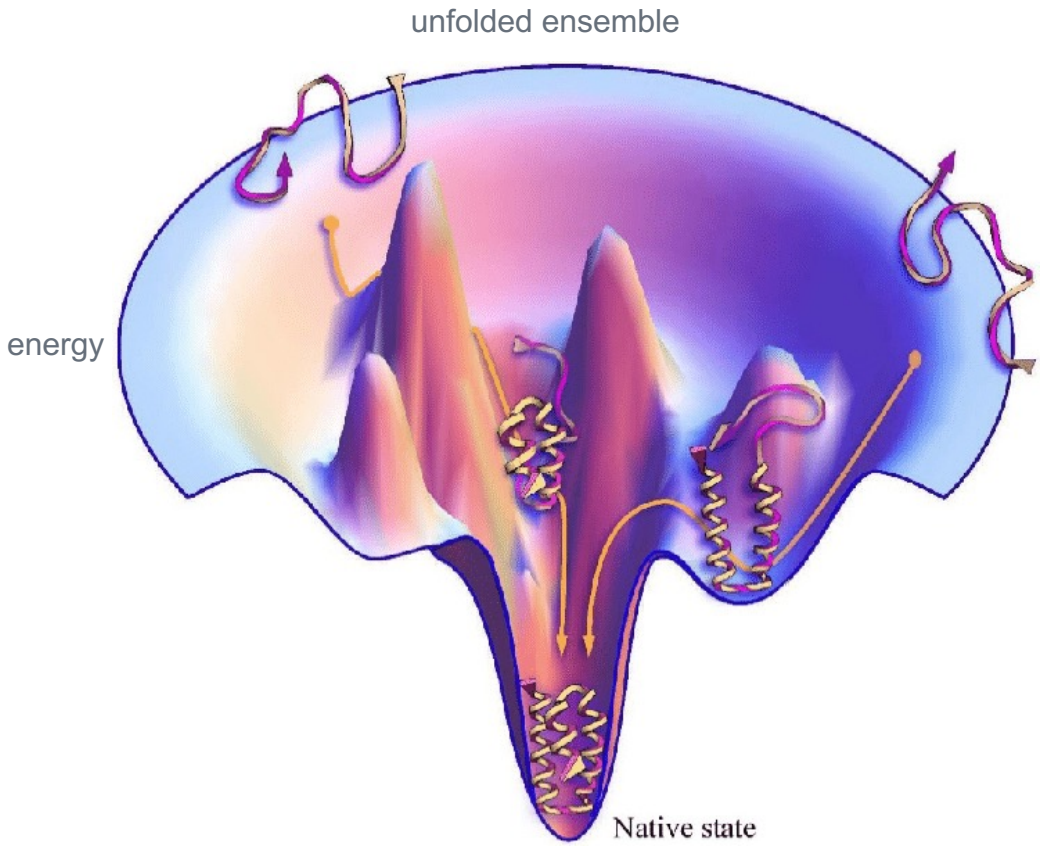
50 AA peptide



http://www.youtube.com/watch?v=fvBO3TqJ6FE

THE RESOLUTION

Folding is a biased descent, not a search



What the funnel buys us

* Local structure forms => fast + restrict
* Funnel towards the native state along many routes
* No enumeration

Why this matters for the rest of today

Native state is encoded in local propensities and pairwise preferences

⬇️

Model that learns these patterns from known structures can get the same answer without the physics.

BEFORE DEEP LEARNING

Fifty years of trying

Physics-based simulation

Molecular dynamics with a force field, folding the chain the way nature does.

Correct in principle, but timescales out of reach

Homology modelling

Copy the backbone of a close homologue and thread the new sequence onto it.

Excellent when template exists. No basis for for a novel fold.

Fragment assembly (Rosetta)

Assemble short fragments from known structures and score the result.

Best template-free option
Accuracy fell off sharply with size and novelty.

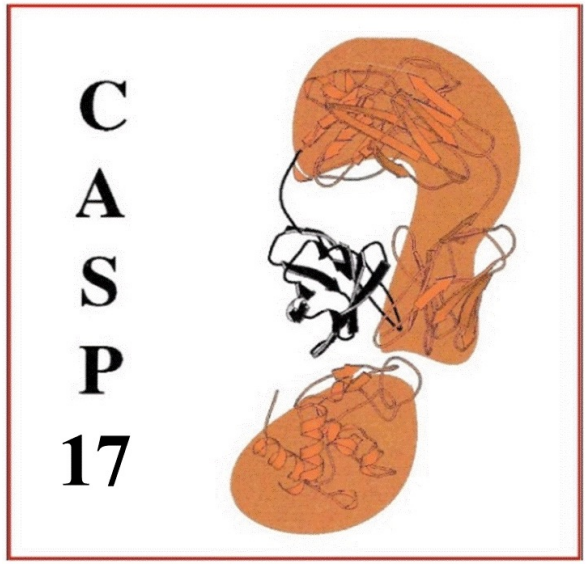
HOW PROGRESS IS MEASURED

CASP: the only honest scoreboard

Blind, by construction

Targets are structures solved but not yet released.

Predictors submit before the experimental coordinates become public, so no method can be tuned to the answer.



Global distance test (GDT)

% of residues that sit within a distance cut-off of their correct position after superposition.

~50 the fold is broadly right, the details are not.

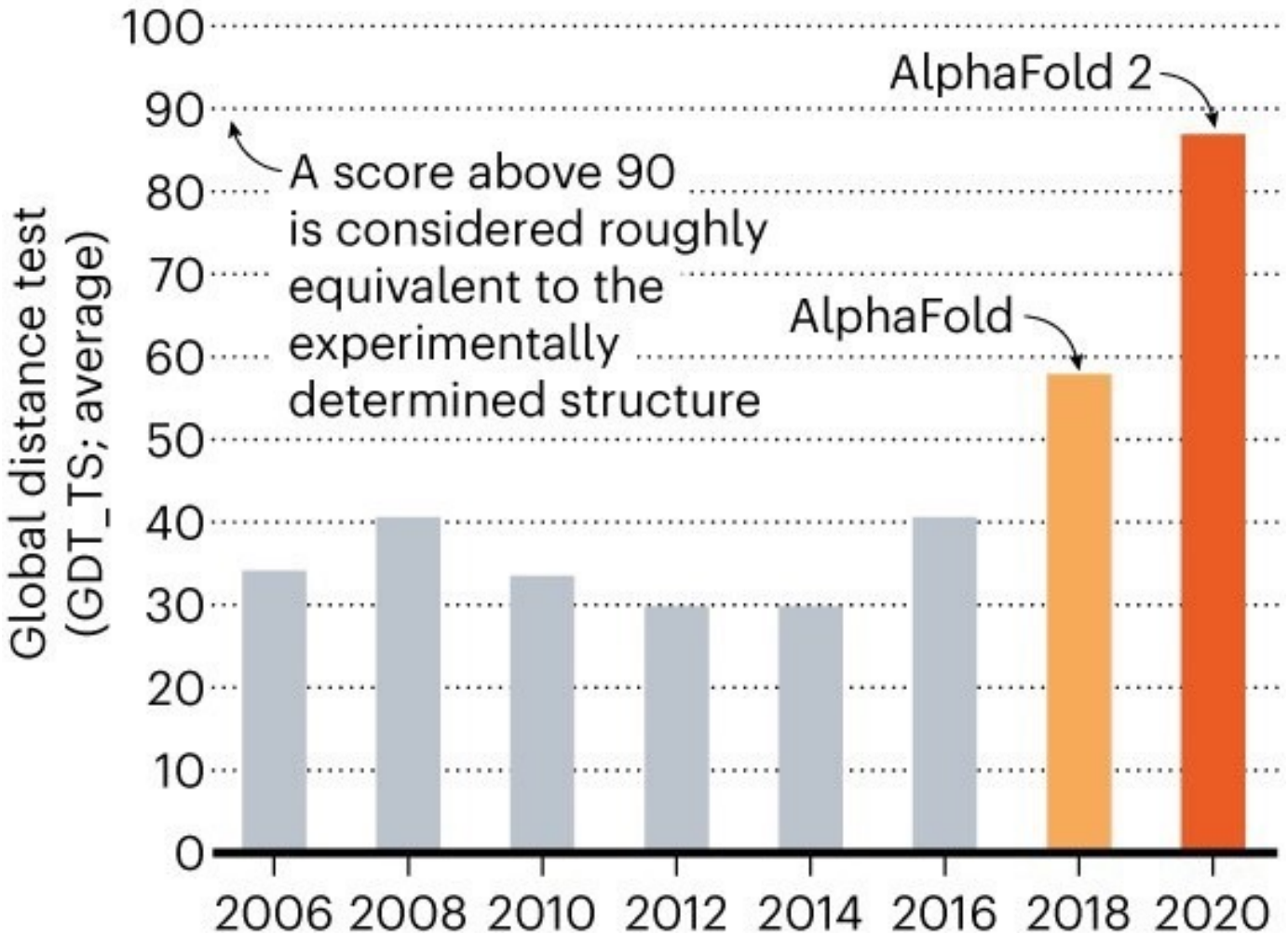
~70 a useful model.

~90 informally regarded as competitive with an experimental structure

https://predictioncenter.org/

CASP PROGRESS

Twenty-four years of grind, then two steps



The experimental line

Around 90 GDT is informally treated as competitive with an experimental structure.

Across all targets

Median 92.4 GDT_TS, with an average error of roughly 1.6 Å, about the width of an atom.

Against the field

Summed z-score of 244.0 for AlphaFold 2, against 90.8 for the next best group at CASP14.

‘It will change everything’: DeepMind’s AI makes gigantic leap in solving protein structures. *Nature* **588**, 203-204 (2020)

INPUTS

What AlphaFold 2 is actually given

Query sequence

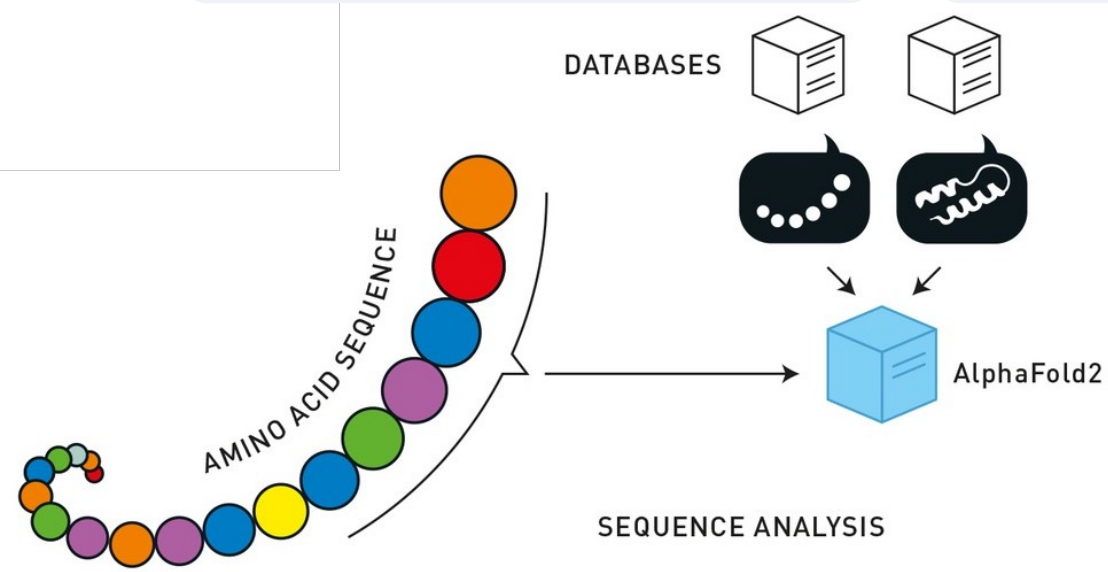
The single chain to be folded.
> Little signal.

Multiple sequence alignment

Homologues collected from the sequence databases.
> Major information source

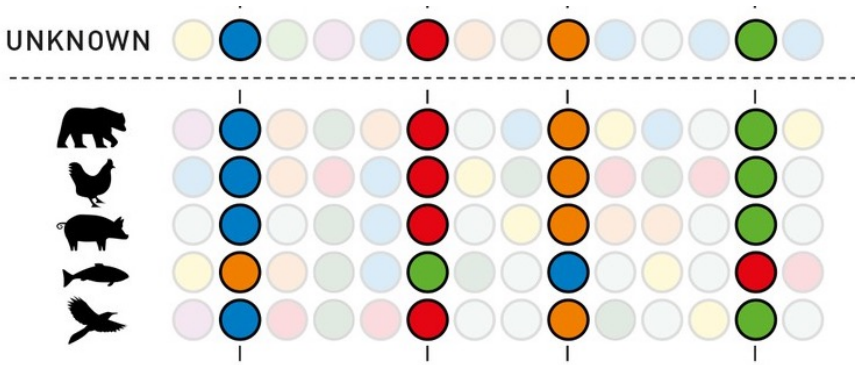
Structural templates

Related structures from the PDB.
> Helpful, but not required.



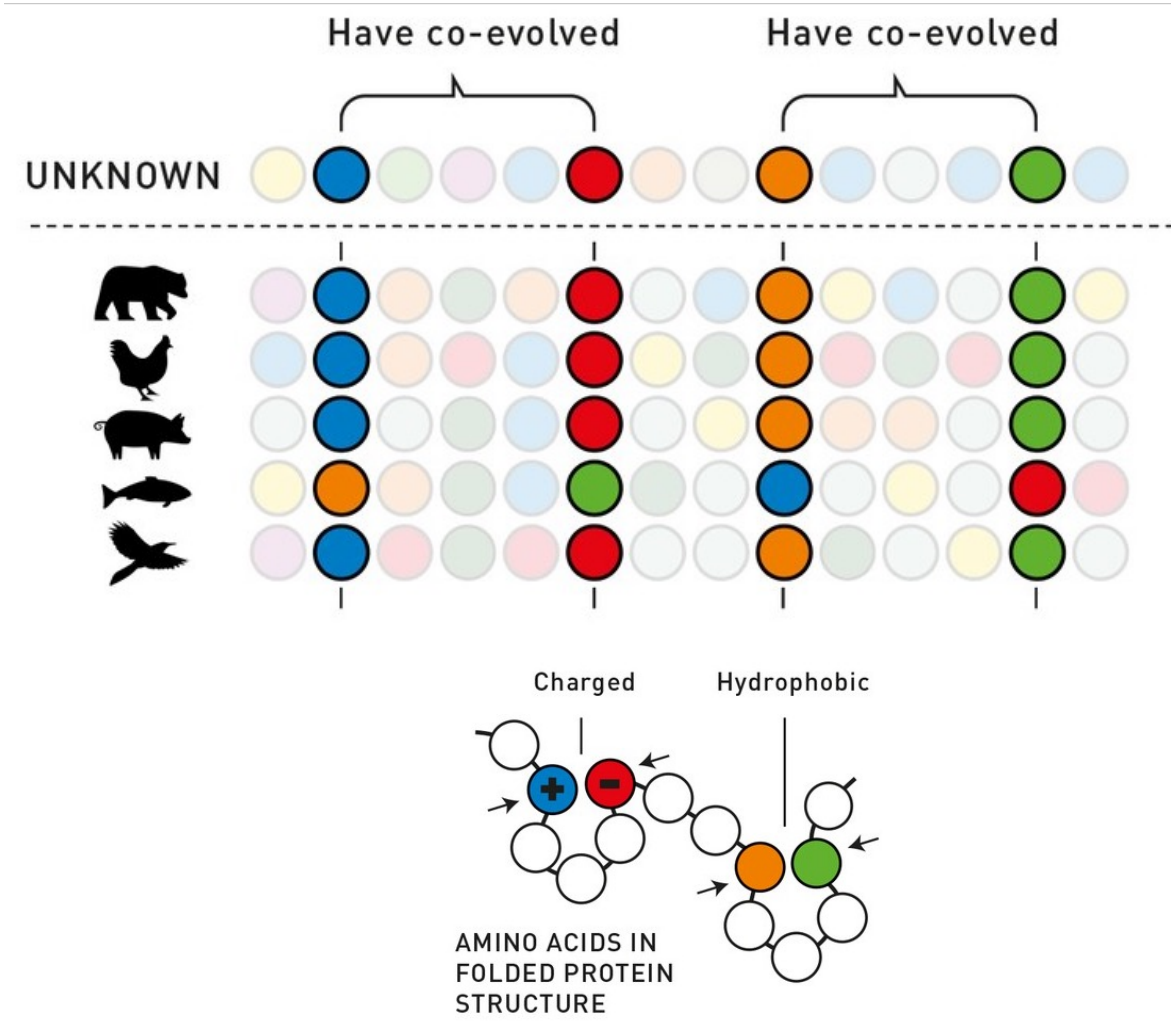
Credit: The Royal Swedish Academy of Sciences

The alignment is the heavy lifter.



WHERE THE SIGNAL COMES FROM

Coevolution: positions that change together



Credit: The Royal Swedish Academy of Sciences

The structural inference

- Two residues in contact ==> shared constraint.
- Change of one often needs a change of the other
- Correlated columns often pairs that are close in space.

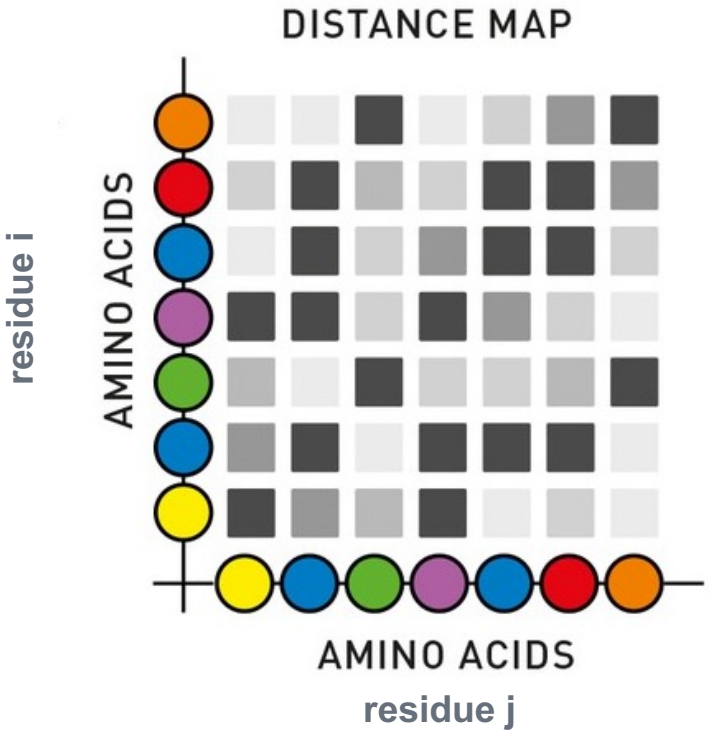
What changed with AlphaFold

Before: couplings were computed explicitly, difficulty was separating direct contacts from indirect, chained correlations.

AlphaFold 2: no explicit coupling step. The network reads the alignment itself and learns what to extract

THE KEY REPRESENTATION

A 3D structure, written as a 2D object



What is predicted

For every residue pair (i, j), a probability distribution over the Cβ–Cβ distance, in roughly 64 interval bins.

Why 2D and not 3D

- No rotation or translation
- Direct matrix input input

Interesting bit

- A complete and consistent set of pairwise distances fixes the 3D structure (+ mirror)
- Lossless conversion.
- What the network learns is the matrix of histograms. The 3D model is the consequence, not the target.
- Confidence scores (later): they are predictions about this matrix.

Credit: The Royal Swedish Academy of Sciences

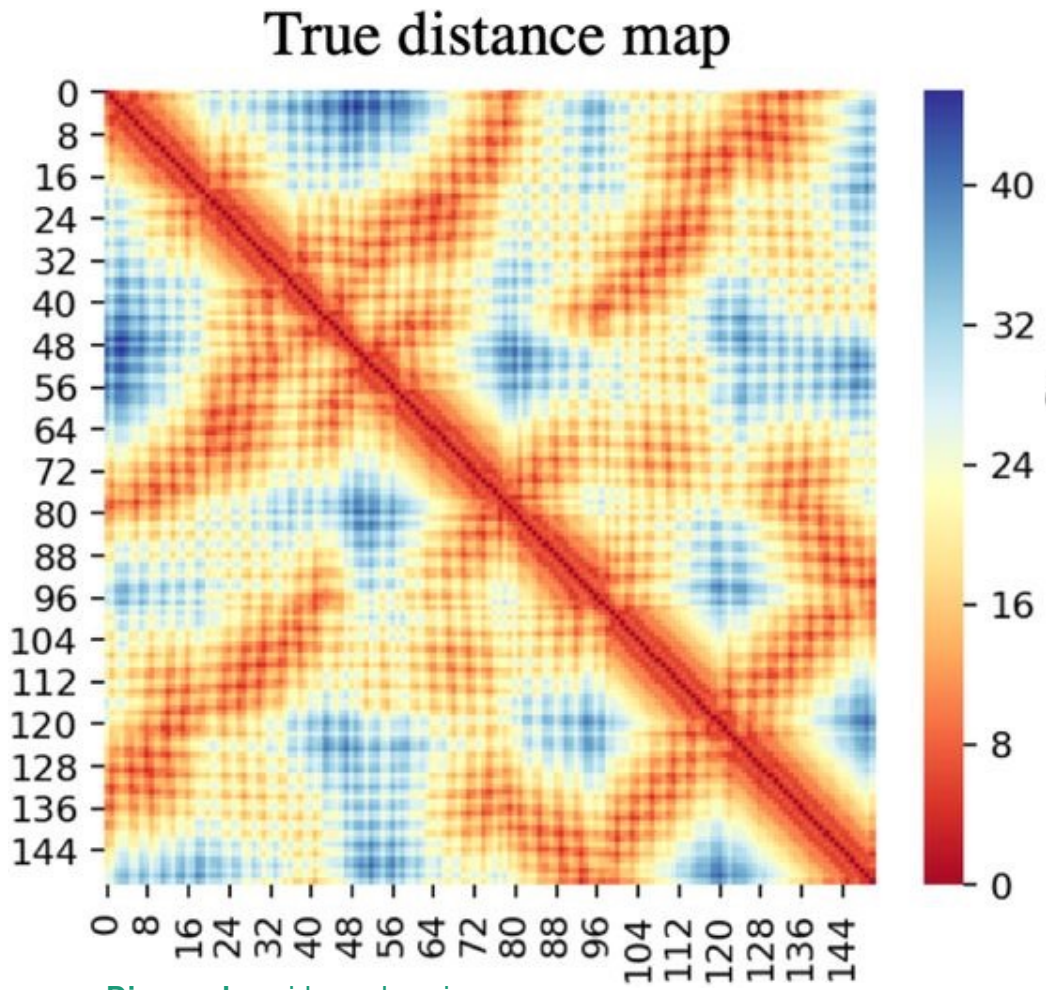
3D structure as a 2D object



Extract

AlphaFold

"Distograms"



Diagonal: residues close in sequence.

Off-diagonal patches: stretches that meet in space.

Jumper et al. (Deepmind), *Nature*, 2021

ARCHITECTURE 1 OF 3

The Evoformer: two views that talk to each other

MSA representation

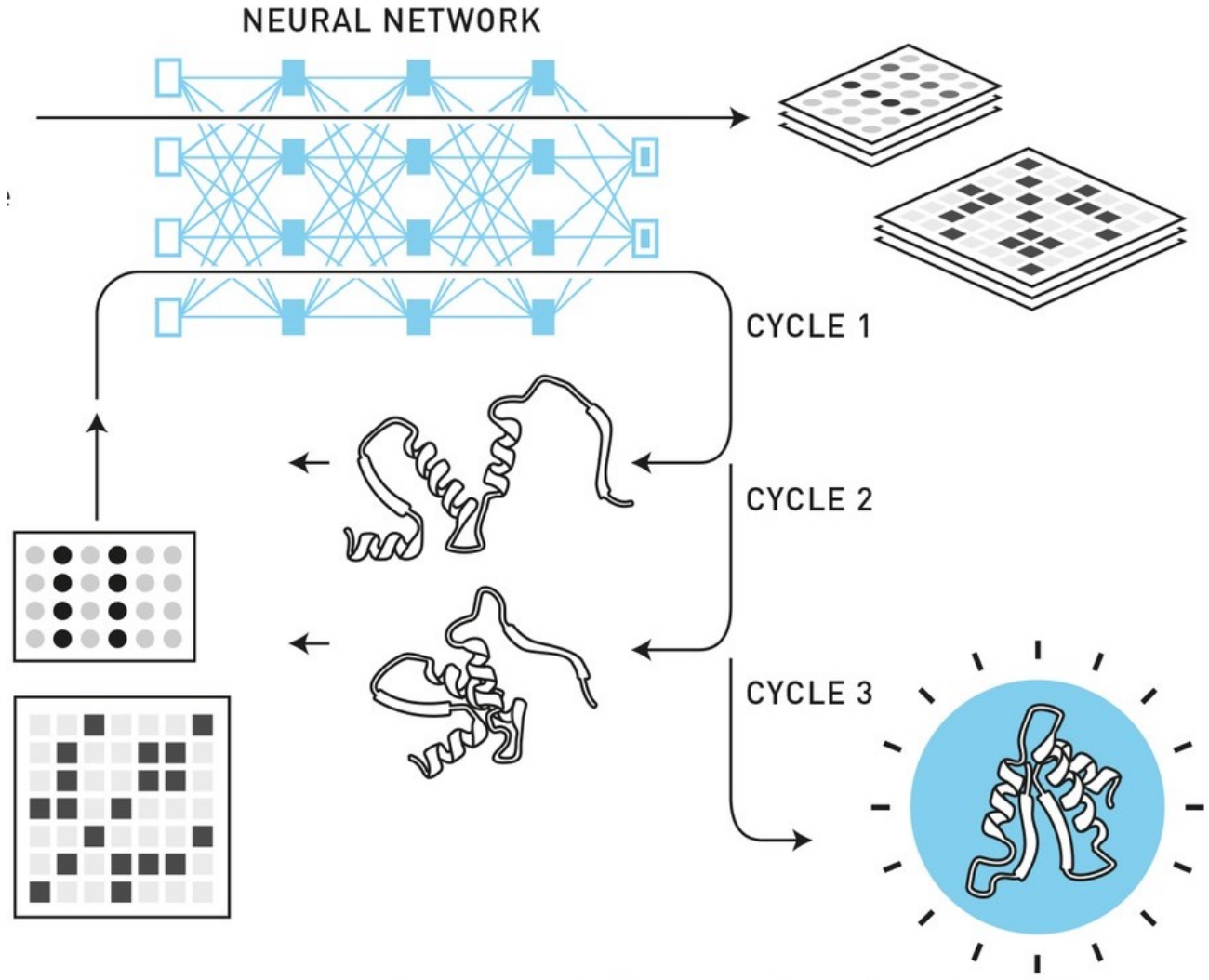
One row per sequence, one column per residue.
What evolution did at each position.

extract

feed back

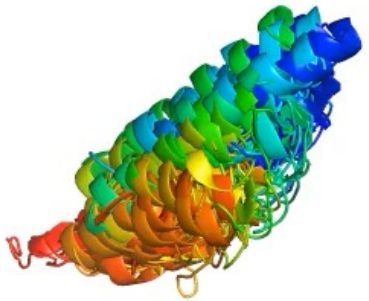
Pair representation

One vector per residue pair (i, j).
This is what becomes the distogram.



EXAMPLES: THE GOOD

When it works, it really works

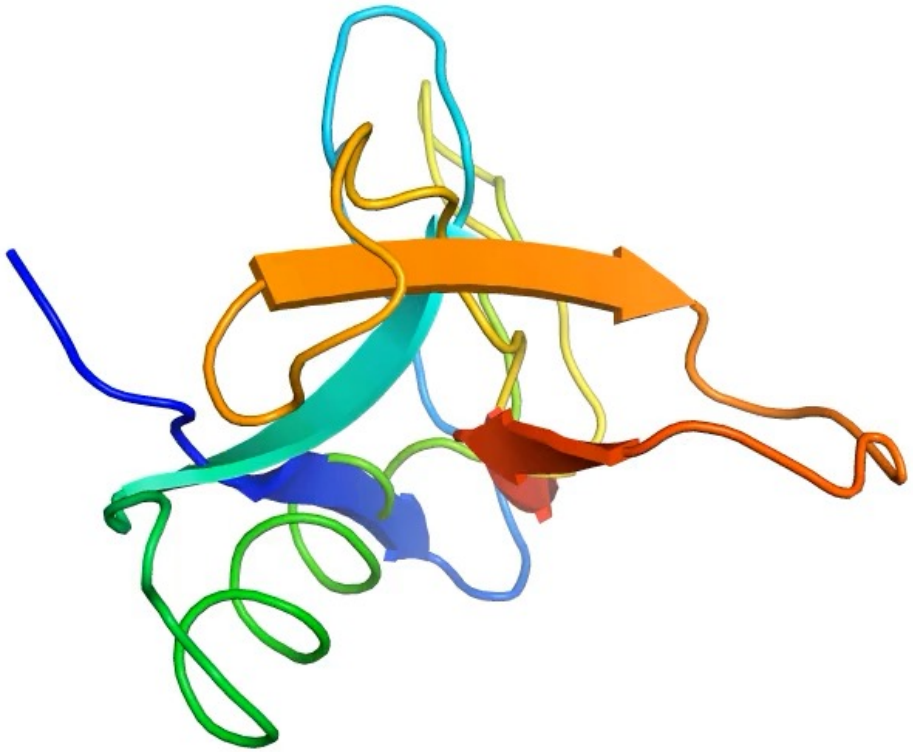


Recycling iteration 0, block 01
Secondary structure assigned from the final prediction

T1044
(RNA polymerase of crAss-like phage)
Large, but easy

EXAMPLES: THE GOOD

When it works, it really works

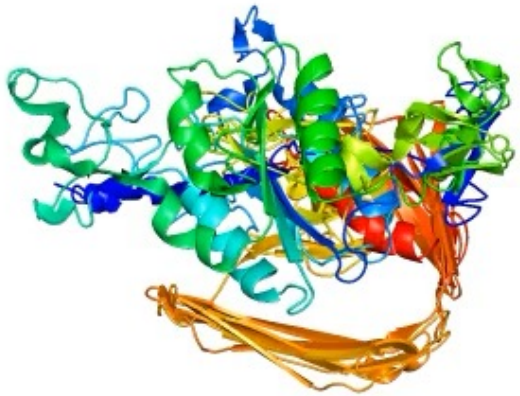


Recycling iteration 0, block 01
Secondary structure assigned from the final prediction

T1064 (Orf8)
A small, but difficult one

EXAMPLES: THE GOOD

When it works, it really works



Recycling iteration 0, block 01
Secondary structure assigned from the final prediction

T1091
A difficult multi-domain example

EXAMPLES: THE BAD

Where it degrades, and why

Shallow alignments

Few homologues - Orphan proteins, fast-evolving families and many viral proteins
Accuracy falls with MSA depth, because the evolutionary information is thin.

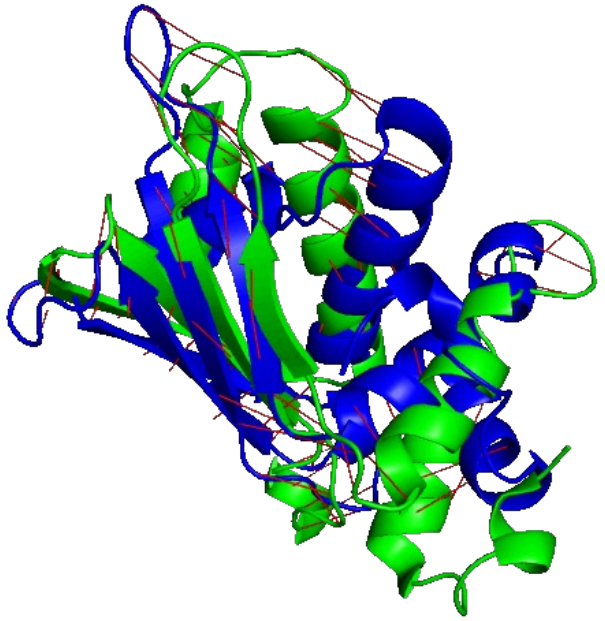
Disordered regions

Regions that have no single structure come back as low-confidence coils or “spaghetti”.

Flexible linkers and multi-domain proteins

Strong domain folds, relative placement is essentially arbitrary.
Single global scores will hide this completely.

None of these is a reason not to use the method. All of them are reasons to read the scores.



Looking ahead

Lesser reliance on alignment depth,

Most failures are detectable from the confidence scores

A SHORT HISTORY

How we got here

1972
Anfinsen
Sequence determines structure

1994
CASP 1
The blind benchmark begins

2011–17
Coevolution
Contact prediction from alignments

2018
AlphaFold
Deep learning wins free modelling

2020
AlphaFold 2
Median 92.4 GDT_TS at CASP14

2021–22
AF DB
200M+ predicted structures released

2022–
OpenFold etc.
Open reimplementation and retraining

Two things changed at once: the predictions became accurate, and then they became public at scale.

RECOGNITION

The Nobel Prize in Chemistry 2024

One half

David Baker

University of Washington

“for computational protein design”

One half, jointly

Demis Hassabis

Google DeepMind

“for protein structure prediction”

One half, jointly

John M. Jumper

Google DeepMind

“for protein structure prediction”



SEGMENT 2

Protein co-folding

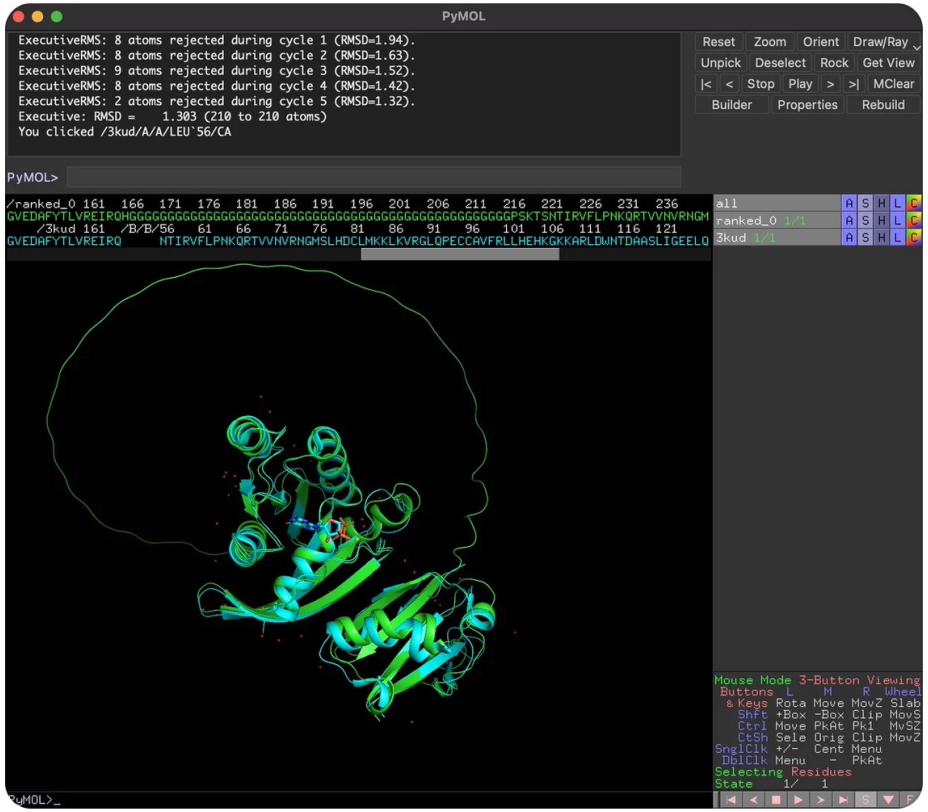
One chain was the benchmark. Two or more is the biology.

CO-FOLDING, THE HARD WAY

The trick that (maybe) started it

Yoshitaka Moriwaki
@Ag_smith

AlphaFold2 can also predict heterocomplexes. All you have to do is input the two sequences you want to predict and connect them with a long linker.

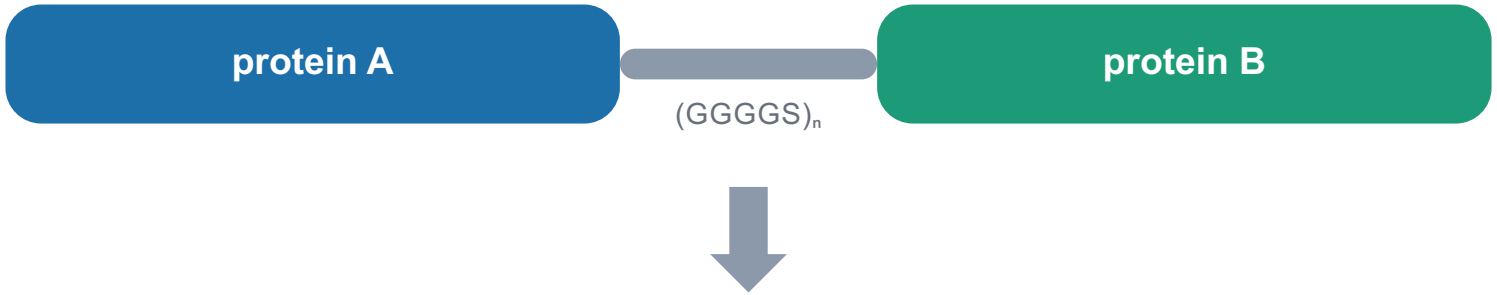


12:07 PM · Jul 19, 2021

CO-FOLDING, THE HARD WAY

The trick that (maybe) started it

One sequence, submitted to a single-chain model



two domains, packed against each other

Why it worked

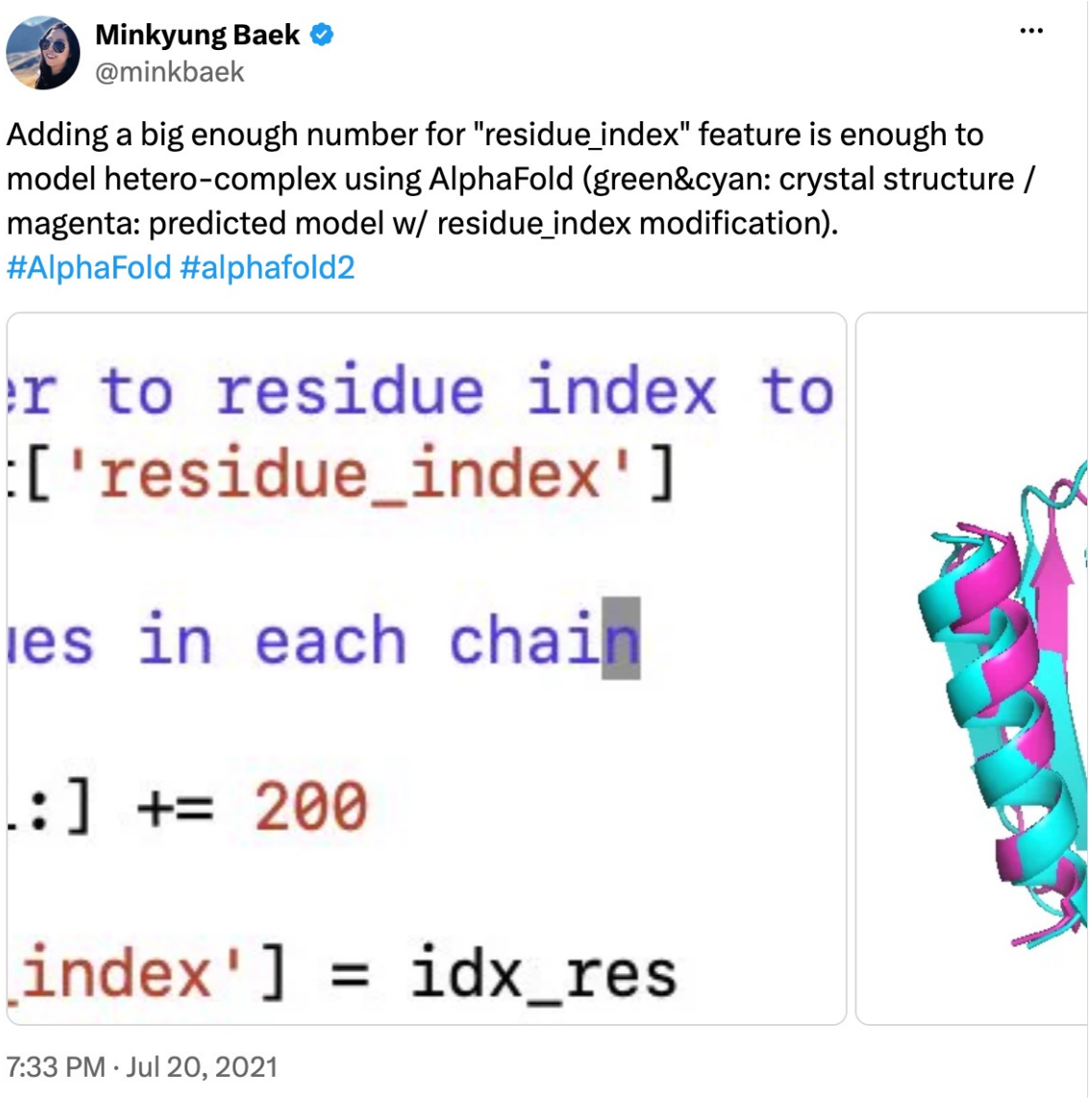
> Long and flexible linker, so the model is free to pack the two halves the way it packs any two domains.

> The single-chain model had already learned about interfaces!

A **hack** rather than a method.
The linker length had to be tuned, it could bias the result, and nothing in the model knew that these were two separate molecules.

CO-FOLDING, SLIGHTLY LESS HARD/BIASED

The same effect, without the linker



CO-FOLDING, SLIGHTLY LESS HARD

The same effect, without the linker

No linker. A jump in the residue numbering instead



What the model sees

A gap in the numbering is read as a break in the chain. The two stretches are treated as effectively independent, and they still fold against each other.

Why this mattered

- linker was never the point. The chain break was.
- Tell the model about chains directly, rather than smuggling the information in through the numbering.

Both tricks say the same thing: the information needed to assemble two proteins was already inside a model that had only ever been trained to fold one.

2021

AlphaFold-Multimer: the trick becomes a method

Trained on complexes

The training set becomes multi-chain assemblies from the PDB rather than isolated chains.

Chains are explicit

Chain identity is an input. The model no longer has to infer a break from a numbering gap.

Paired alignments

Homologues of both partners are paired by organism, so coevolution across the interface becomes visible.

Interface-aware scoring

Confidence is reported for the interface specifically, not only for the fold.

2024

AlphaFold 3: the structure module is replaced

AlphaFold 2

Evoformer over MSA and pair views, then a structure module that places residue frames.



AlphaFold 3

A smaller MSA module feeding the Pairformer, then a diffusion head that generates atomic coordinates directly rather than placing residue frames.

Less weight on the alignment

The MSA stack shrinks and the pair representation carries more of the load. One of several attempts across the field to depend less on alignment depth.

Atoms, not residues

The diffusion head generates atoms. Nothing in it assumes an amino acid, which is exactly why the molecule types on the next slide become possible.

WHAT THIS OPENS UP

One model, more than one kind of molecule

Proteins

single chains and complexes

Small molecules

ligands in their pocket

DNA and RNA

nucleic acid complexes

Ions and PTMs

cofactors, metals, modifications

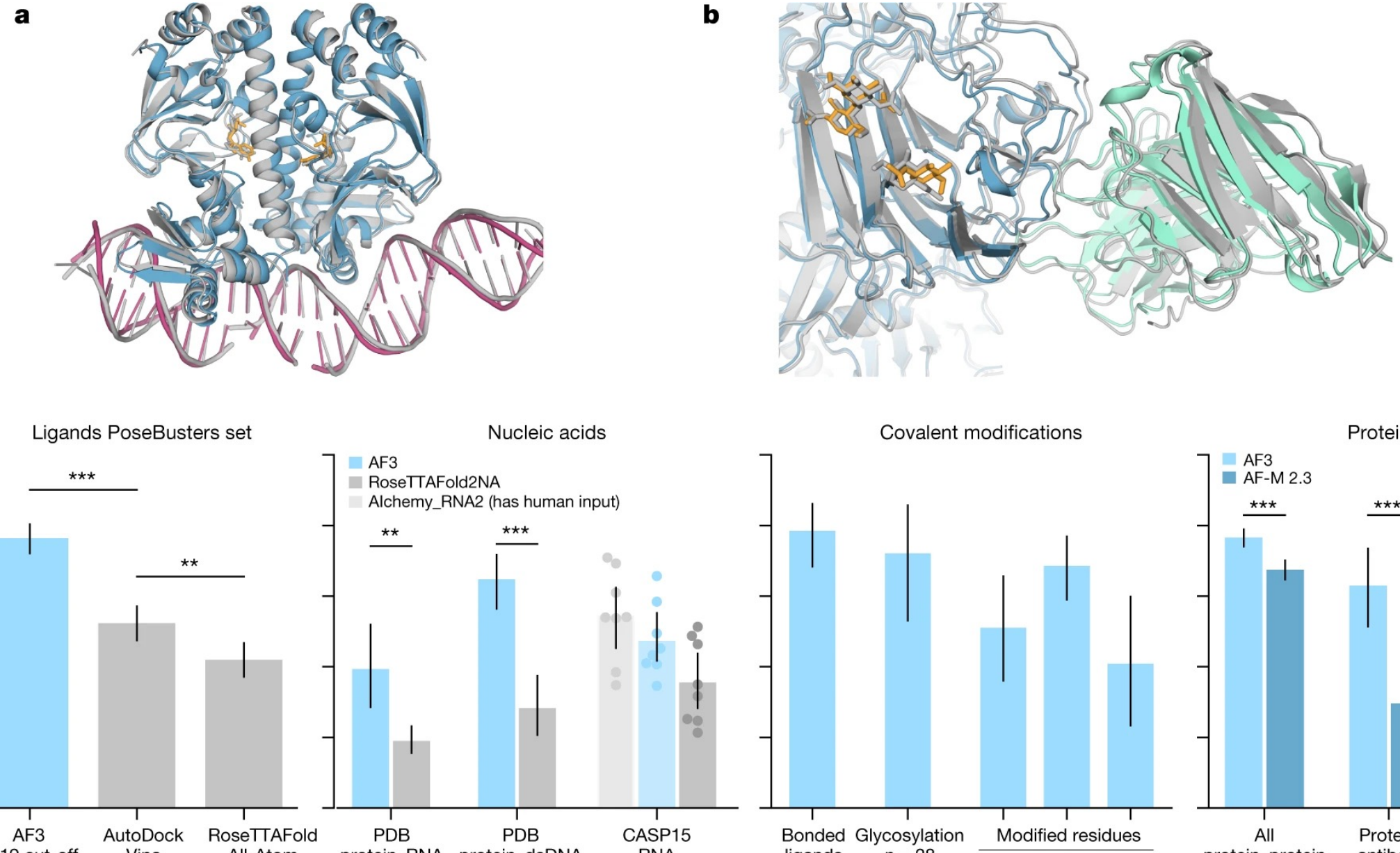
And the honest caveats

Performance varies a great deal between systems.

Ligand poses are not uniformly of docking quality.

Every confidence caveat from the next section still applies.

WHAT THIS OPENS UP



AI- AND DATA-DRIVEN DRUG DESIGN · LECTURE 7

How confident is the model?

pLDDT, PAE, pTM, ipTM and ipSAE, built up step by step

CONFIDENCE METRICS

Now the part you will actually use

Every(!) one of these methods returns a structure.

A prediction always looks like a protein.

The only way to answer it without the experimental structure is to read the confidence output properly.

pLDDT

Is this region placed well locally?

PAE

Are these two parts placed correctly relative to each other?

pTM / ipTM

Is the overall fold, or the interface, trustworthy?

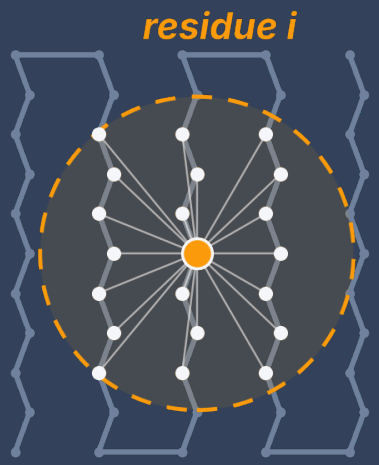
ipSAE

Is this specific interface real, independent of chain size?

LOCAL CONFIDENCE

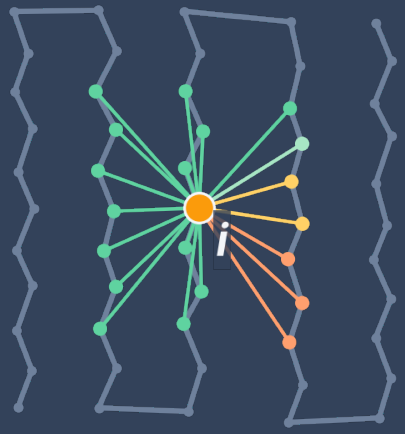
lDDT: are local distances preserved?

True structure



20 neighbours within 15 Å

Predicted model



Same pairs re-measured: green = kept, red = lost

Fraction of distances kept



lDDT of residue i = 0.82

mean of the four fractions

pLDDT is the network's own prediction of this score for every residue, on a 0–100 scale.

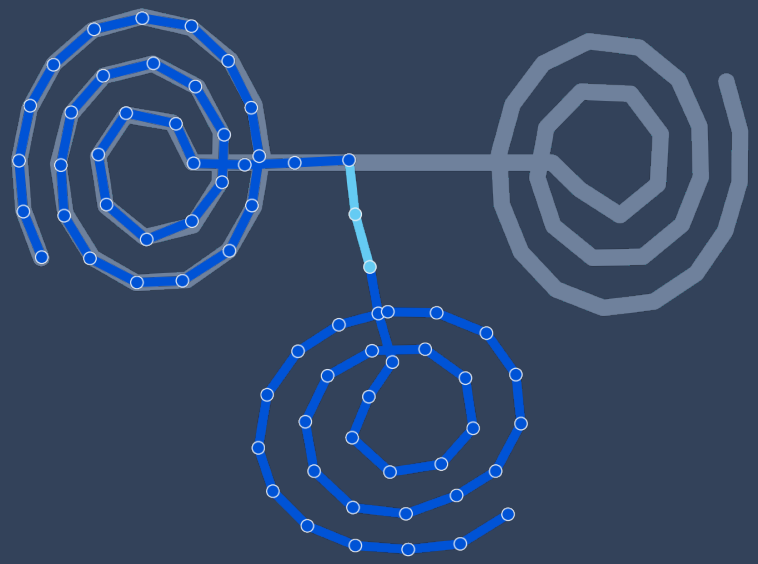
above 90: very high | 70–90: confident | 50–70: low | below 50: very low

Mariani et al., Bioinformatics 2013 · Jumper et al., Nature 2021 · Illustration: 2-D toy model; all numbers are computed from sampled models

LOCAL CONFIDENCE

pLDDT: what can it not see?

True structure: two domains and a hinge



≈ 7 Å

global Cα-RMSD after the best superposition

99

mean lDDT-Cα: every 15 Å neighbourhood is intact

Model, coloured by per-residue lDDT: one domain has swung away

High pLDDT everywhere does not mean the domains are placed correctly. That needs a pairwise score.

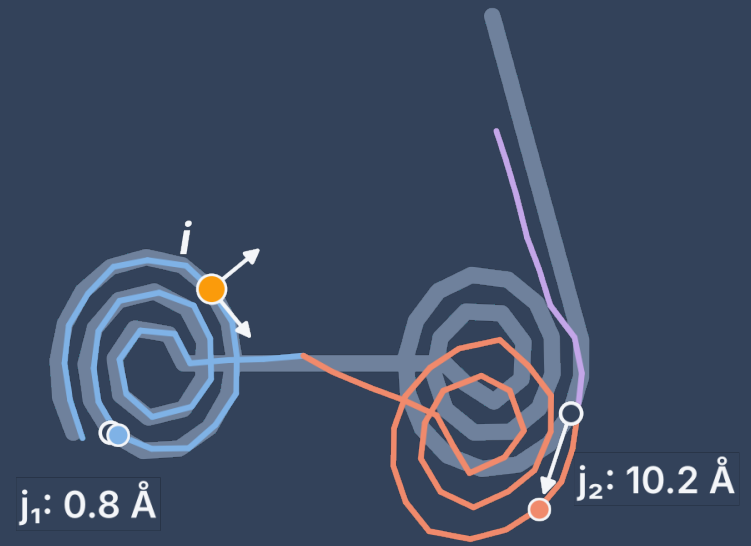
Illustration: 2-D toy model; all numbers are computed from sampled models

PAIRWISE CONFIDENCE

PAE: where is j, seen from i?

Align model and truth on residue i

grey = true structure, colour = model



One row = one alignment

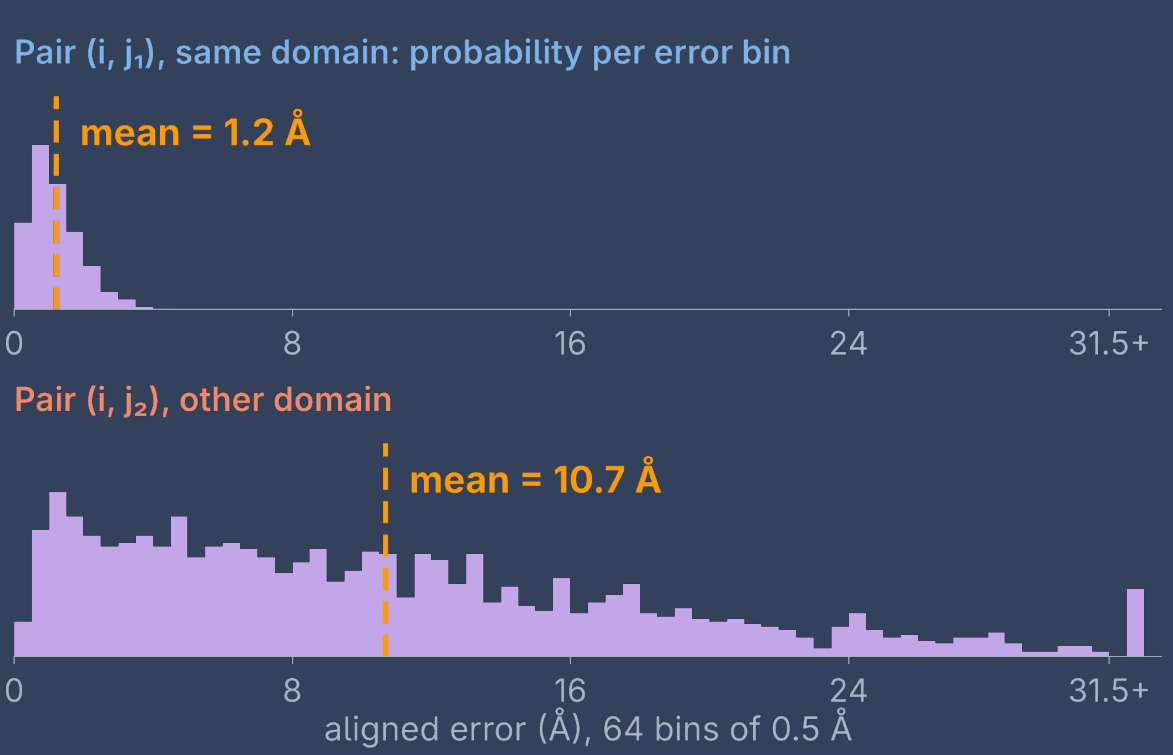


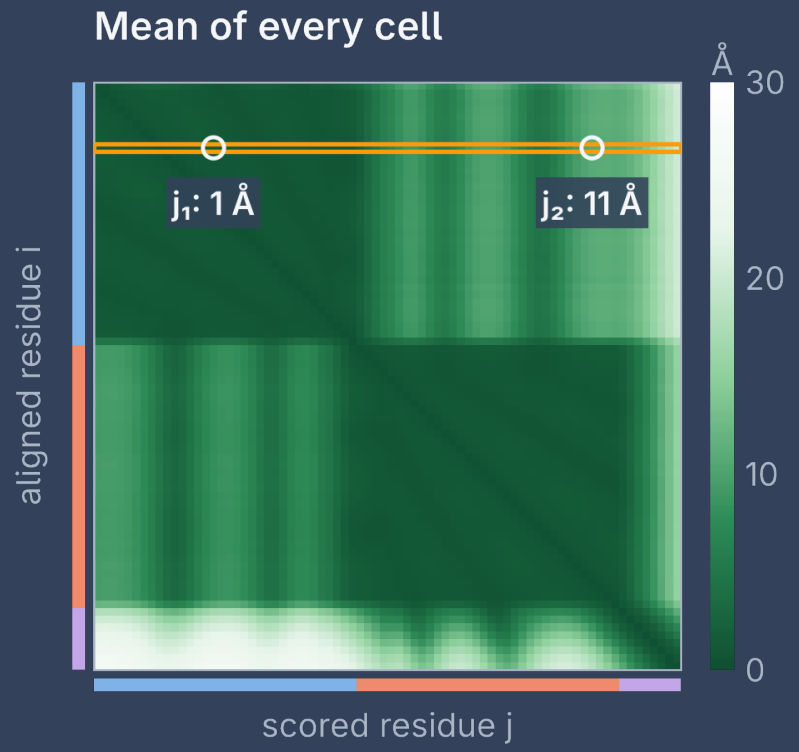
PAE(i, j): expected error at j after aligning on i. One row per alignment, so it is not symmetric.

Illustration: 2-D toy model; all numbers are computed from sampled models

PAIRWISE CONFIDENCE

PAE: how does the network predict it?





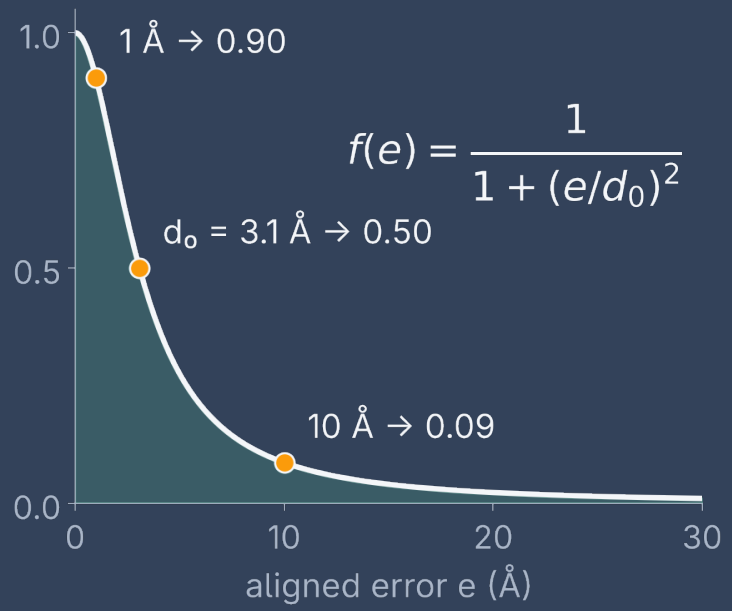
Trained on true aligned errors; no true structure is needed at inference.

64 bins of 0.5 Å, last bin open-ended · AF2: pair-representation head · AF3: confidence head · 2-D toy model, numbers computed

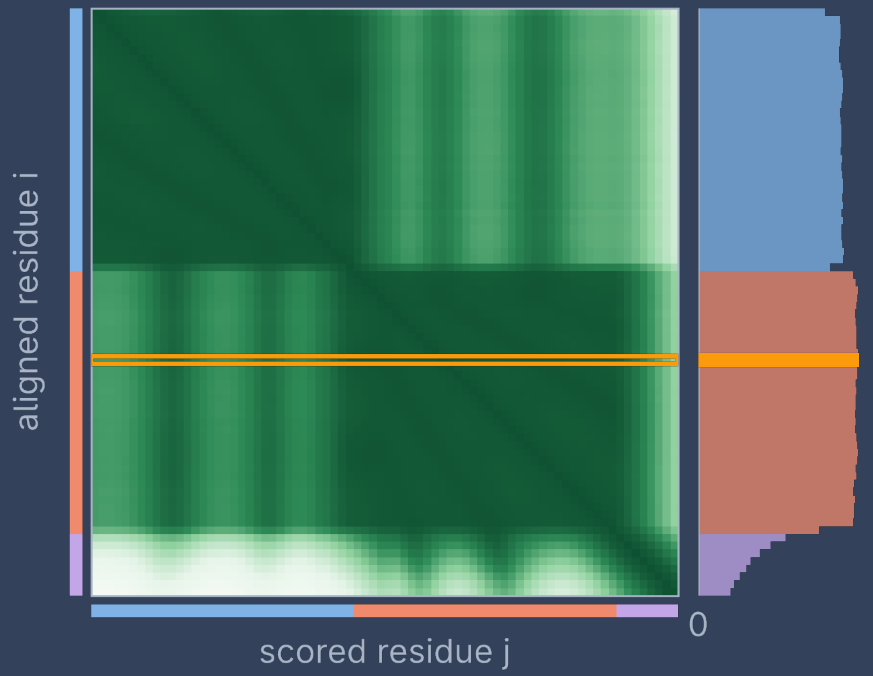
GLOBAL SCORE

pTM: how good is the best superposition?

1 · Score each pair



2 · Mean per row



3 · Take the max

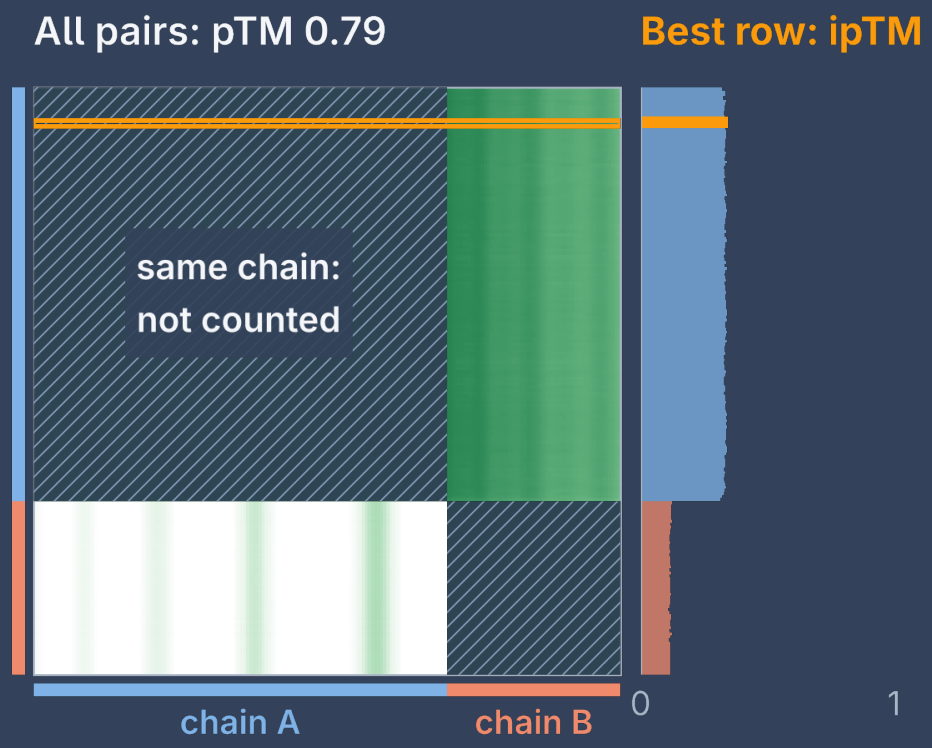
pTM = 0.63

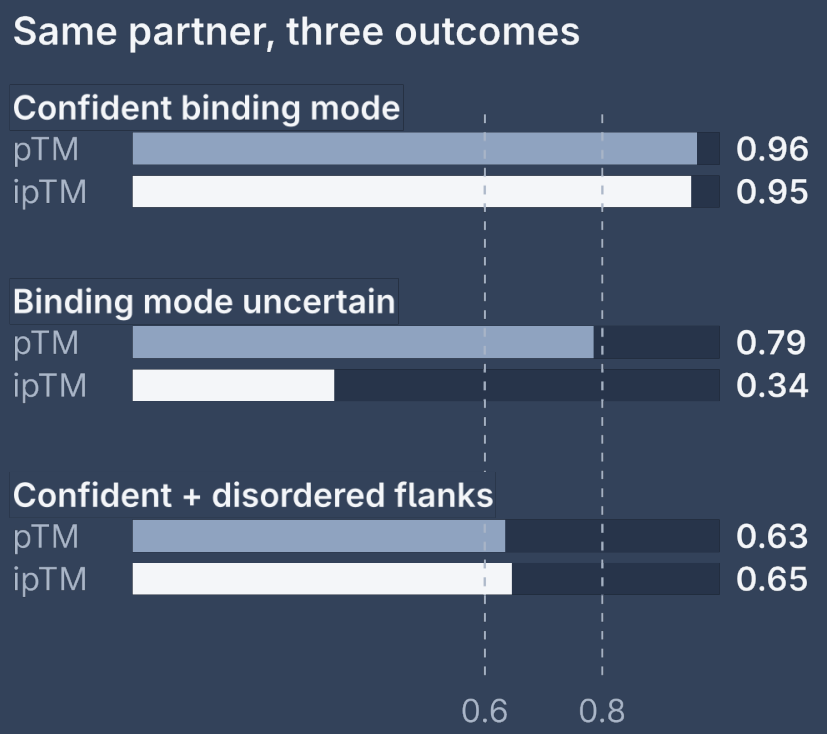
Each row is a candidate superposition; the max keeps the best anchor. Every scored residue in that row still counts.

Zhang & Skolnick, Proteins 2004 · Jumper et al., Nature 2021 · Illustration: 2-D toy model; all numbers are computed from sampled models

INTERFACE SCORE

ipTM: are the chains placed correctly?





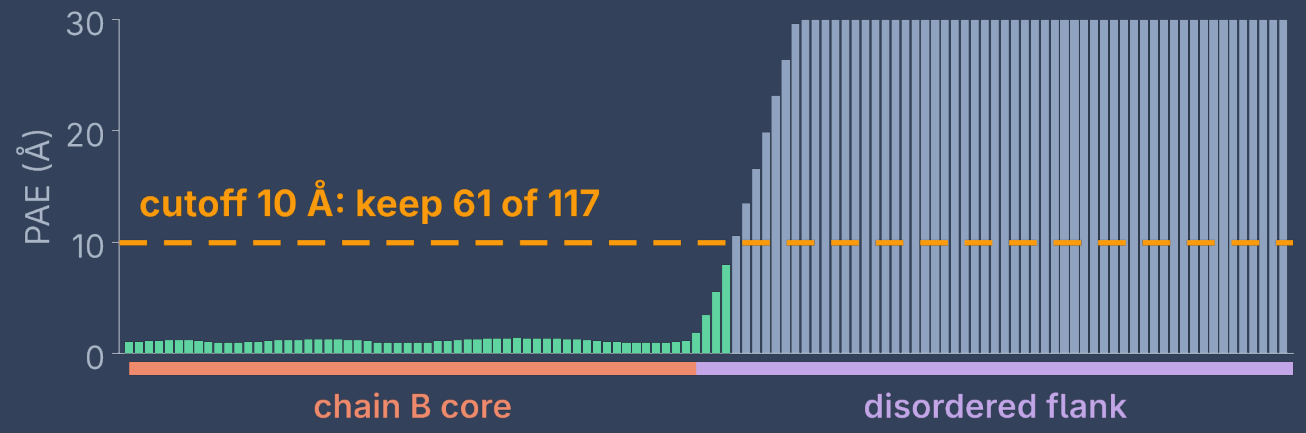
Above 0.8 confident, 0.6–0.8 grey zone, below 0.6 likely failed.
All partner residues count, so disorder dilutes it.

Evans et al., bioRxiv 2021 · AF3 ranking: 0.8 ipTM + 0.2 pTM + 0.5 disorder − 100 has_clash · 2-D toy model, numbers computed

INTERFACE SCORE

ipSAE: what if only confident pairs count?

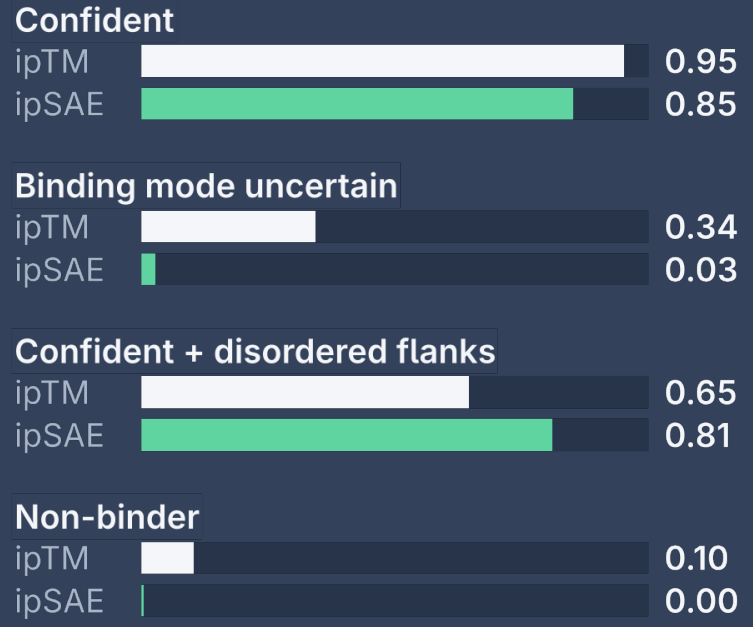
One row: residue i in chain A against all of chain B



ipTM-style: all j, d_0 from N = 313 — 0.51

ipSAE: kept j only, d_0 from those 61 — 0.81

ipTM vs ipSAE, four cases



Same max over rows, but only confident inter-chain pairs count.
No standard thresholds: calibrate on known binders.

Dunbrack 2025 · ipsae.py default cutoff: 10 Å (AF3, Boltz), 15 Å (AF2) · Illustration: 2-D toy model; all numbers are computed from sampled models

SUMMARY

Five scores, five questions

SCORE	PER	QUESTION IT ANSWERS	WATCH OUT
pLDDT	residue	Is this residue's local environment right?	Says nothing about domain or chain placement
PAE	residue pair	Is j placed correctly relative to i?	Not symmetric; the plot shows only the mean
pTM	whole prediction	Is the overall fold or assembly right?	Can look fine when the interface is wrong
ipTM	interface	Are the chains placed correctly relative to each other?	Diluted by disordered regions
ipSAE	chain pair	Is there a confident interface, counting only confident pairs?	No standard thresholds; calibrate on known binders

SEGMENT 3

Protein design

FROM PREDICTION TO DESIGN

Inverting the question

Prediction

sequence → structure

One native state. The answer is effectively unique.

Design

structure or function → sequence

Massively degenerate. Very many sequences give the same fold.

The degeneracy is the point

Many sequences fold to the same shape
Find one that is also stable, expressible and specific.

What has to be generated

A backbone that could exist, and a sequence that folds into it. Current methods either do these in two steps or, more recently, together.

THE BASELINE

Design before generative models

1

Start from a backbone you already have

Either a natural scaffold or one assembled by hand from idealised secondary structure elements.

2

Search sequence space against an energy function

Monte Carlo over rotamers and sequence, scored with a physics-based potential.

3

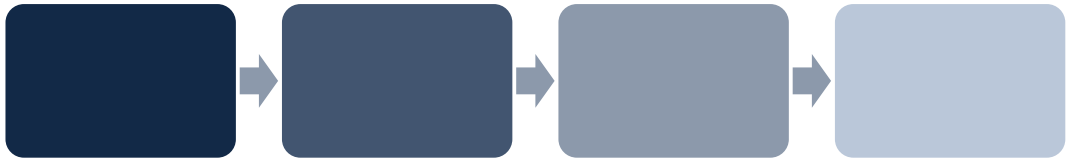
Real successes, real limits

De novo folds and designed binders were achieved, but every design began from a backbone that somebody had to supply.

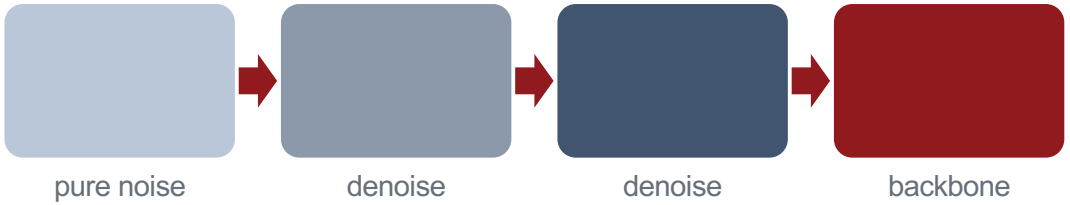
THE GENERATIVE IDEA

Diffusion: learn to remove noise, then start from pure noise

Training: add noise, learn to remove it



Generation: run it backwards



This is the same mechanism that replaced the structure module in AlphaFold 3. One idea, doing work on both halves of today's lecture.

Why start from noise at all

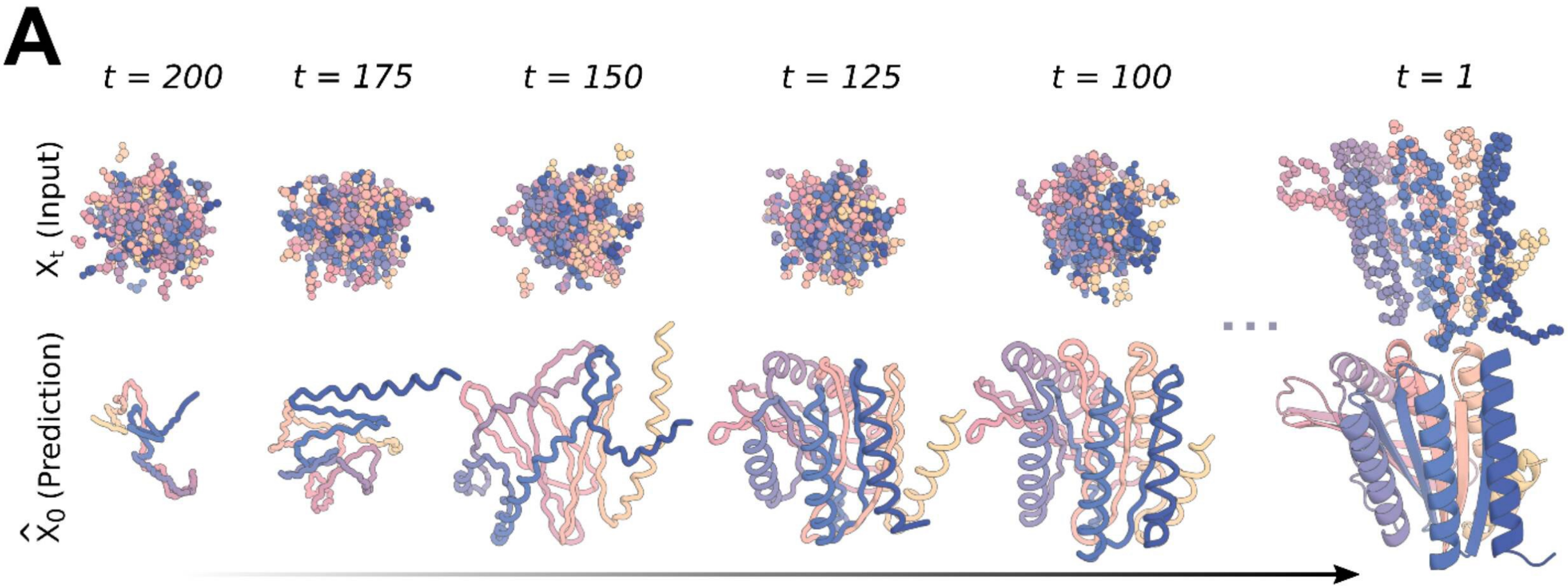
Because the starting point is random, every run produces a different valid answer. That is what you want in design, and it is exactly what you would not want in prediction.

Why it is controllable

The denoiser can be conditioned. Hold part of the system fixed, such as a target protein and the residues you want contacted, and let the model generate only the rest around it.

THE GENERATIVE IDEA

Diffusion: learn to remove noise, then start from pure noise



https://www.biorxiv.org/content/10.1101/2022.12.09.519842v2

BACKBONE GENERATION

RFdiffusion: diffusion conditioned on a target

Target

The protein you want to bind, held fixed throughout.

Hotspots

Optionally, residues on the target surface you want the binder to contact.

Length

How long the designed chain should be.

Diffusion

A backbone is denoised into place against the fixed target.

The same machinery, several tasks

Binder design: generate a chain that packs against a chosen surface.

Motif scaffolding: hold a functional motif fixed and build a protein around it.

Symmetric assemblies: generate oligomers under a symmetry constraint.

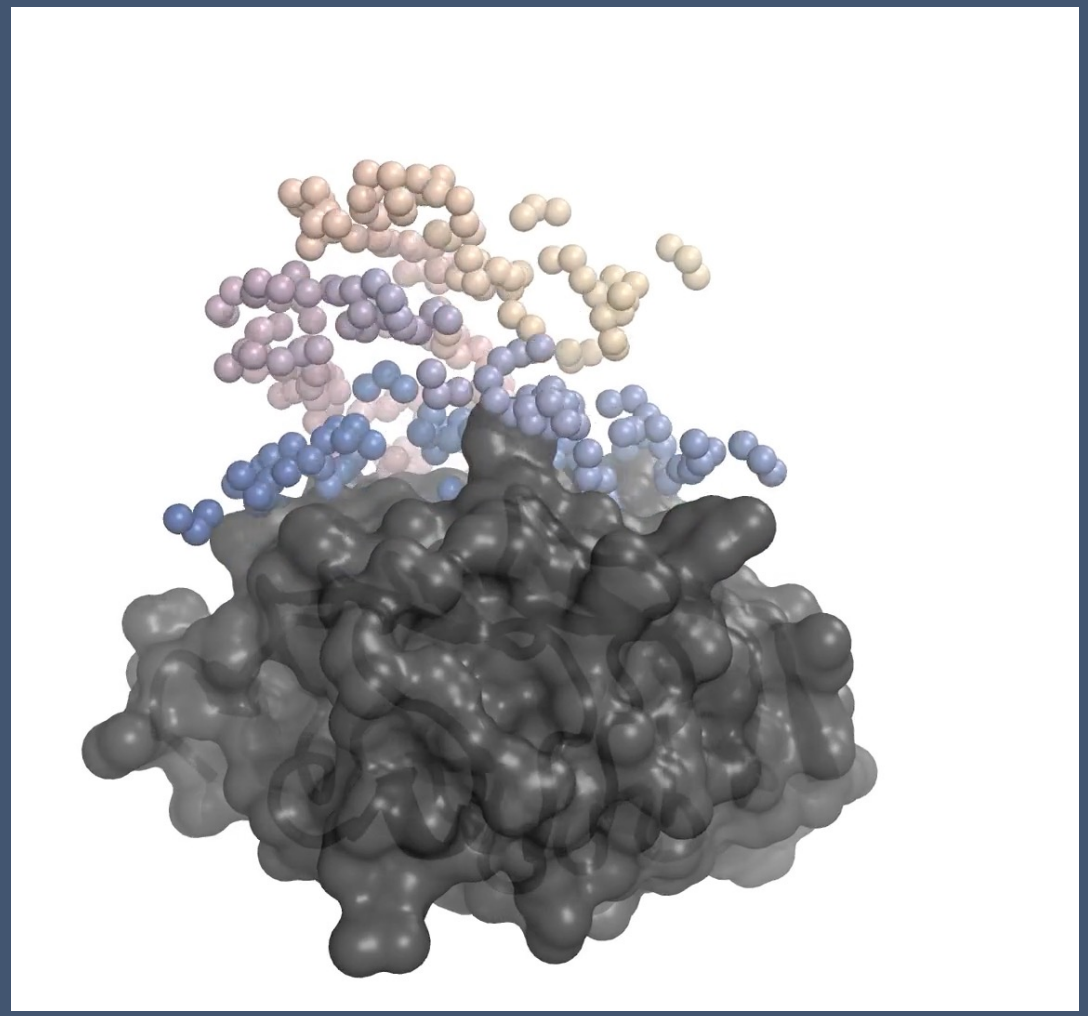
Unconditional generation: new folds, with no target at all.

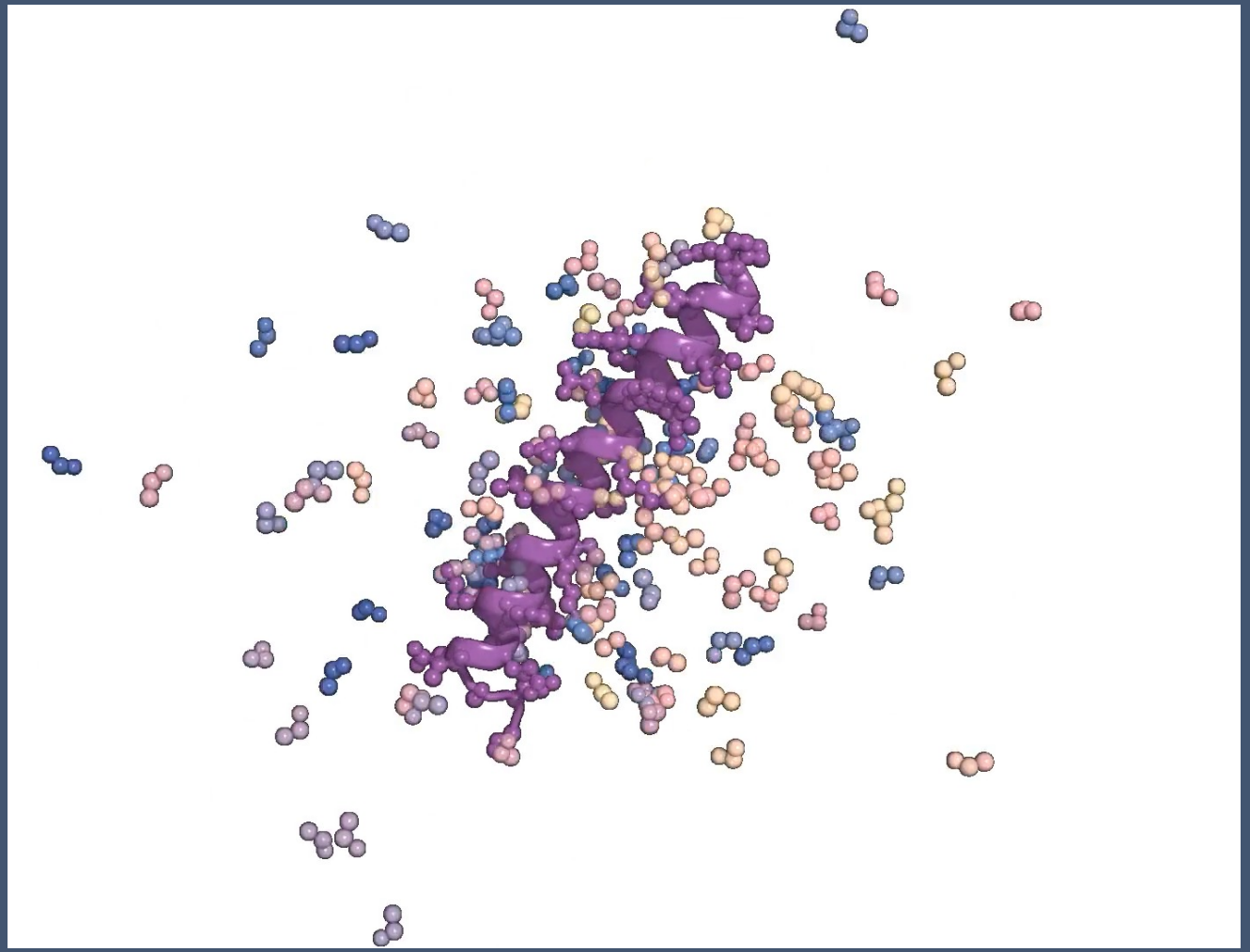
What comes out

A backbone. Atoms along the main chain, in a shape that could exist and that fits the target. There is no sequence yet, and that is the subject of the next three slides.

RFDIFFUSION IN MOTION

Watching a binder appear





https://www.bakerlab.org/2022/11/30/diffusion-model-for-protein-design/

THE GAP

A backbone is not yet a protein

You have a shape.

You do not have a sequence.

RFdiffusion returns main-chain atoms. No position carries an amino acid identity, so there is nothing to order, nothing to express and nothing to test.

The question becomes: which sequence folds into this backbone? That is inverse folding.

INVERSE FOLDING

ProteinMPNN: a sequence for a given backbone

How it works

The backbone is treated as a graph: residues are nodes, and the geometric relationships between neighbouring residues are edges. The network predicts amino acid identities conditioned on that geometry.

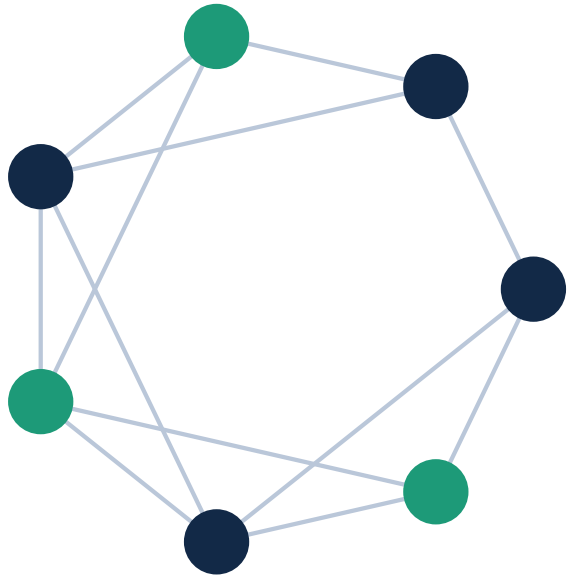
In practice

Fast, and far more reliable than the physics-based fixed-backbone design it replaced.

Samples sequences, so many designs can be generated per backbone at a chosen temperature.

Positions can be held fixed where a motif has to be preserved.

→ M K V L I A E …



backbone as a graph of residues

AN IMPORTANT CAVEAT

What ProteinMPNN is optimising, and what it is not

What it optimises

Compatibility between a sequence and a backbone. Will this sequence fold into this shape?

What it does not optimise

The quality of the interface with your target. Those side chains were chosen for the fold, not for the contact.

So the interface has to be evaluated separately.

1. Generate

RFdiffusion backbones against the target

2. Design

ProteinMPNN sequences for each backbone

3. Refold

Predict the designed complex with AlphaFold or an equivalent

4. Filter

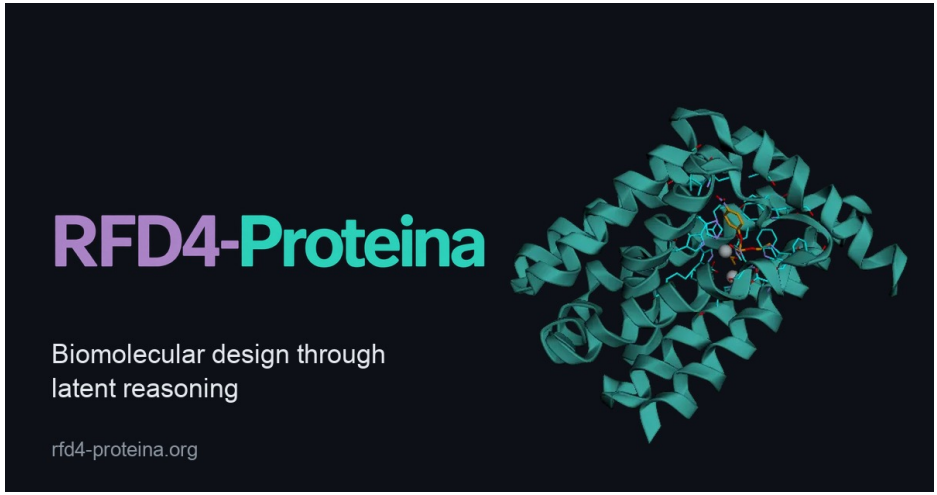
Read pLDDT, PAE and ipSAE at the interface, and keep the survivors

OCTOBER 2026

RFD4-Proteina: sequence and structure, designed together

The change that matters

Sequence and all-atom structure are generated jointly, in one model. The two-step split between backbone generation and inverse folding collapses into a single step.



Who Baker Lab, Institute for Protein Design (UW), with NVIDIA GenAIR

When Public release 19 October 2026

How Flow matching rather than diffusion in the strict sense.

What it covers Proteins, nucleic acids and small molecules. Binder design, motif scaffolding, enzyme design, antibody design, inverse folding

STATE OF THE ART

A crowded and fast-moving field

Prediction, proteins

AlphaFold 2
the reference

OpenFold
open reimplementation, retrainable

ESMFold
language-model based, no MSA

Prediction, complexes and ligands

AlphaFold 3
the reference

Chai-1
open implementation

Protenix
open implementation

Boltz
open implementation

OpenFold 3
open reimplementation, retrainable

OpenDDE/Mol-1/TorchFold/...

Design

RFdiffusion
backbone generation

ProteinMPNN
inverse folding

RFD4-Proteína
joint sequence and structure

Boltz-Design/PxDesign/ColabDesign/…
design on top of predictors

WORKSHOP 3

What you will run this afternoon

1

Fold

AlphaFold / OpenFold

Run a prediction, then read pLDDT and the PAE matrix before looking at the structure.

2

Co-fold

A complex of your choice

Decide whether the interface is real, using ipTM and ipSAE rather than first impressions.

3

Design

RFD4-Proteína

Generate binders against a target, then filter them and defend what survives.

At every stage, the question is not whether you got a structure. You will always get a structure. The question is whether you can defend keeping it.

SUMMARY

Four things to take away

1 The representation is 2D

AlphaFold learns a matrix of pairwise distance distributions. The 3D structure is the consequence, not the target.

2 Co-folding started as a trick

A linker, then a numbering gap, then an architecture. Most of what was needed had already been learned from single chains.

3 The metrics are the usable part

pLDDT is local and separate. pTM, ipTM and ipSAE all come out of the PAE. No single number answers the whole question.

4 Design inverts the problem

Generate a backbone, design a sequence, or now do both at once. Because every step is generative, filtering is part of the method.

If you remember one slide from today, make it the table of which metric answers which question.